

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0675

Halldale Tap 138 kV
October 22, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0675. The Interconnection Customer proposes a 241.8 MW wind facility interconnecting at the Halldale Tap 138 kV substation in the Pennsylvania area. The request was received on September 10, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 316 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,650,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Enka - Houghton Rock 138kV	Hooversville - Halldale Tap 138kV line	102.3	191	66.7
Houghton Rock - Halldale Tap 138kV	Halldale Tap - Hooversville 138kV	100.2	200	66.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Enka - Houghton Rock 138kV Line	Reconductor line	\$1,063,000	100% (\$1,063,000)
Houghton Rock - Halldale Tap 138kV	Reconductor line	\$1,587,000	100% (\$1,587,000)
	Total allocated to project		\$2,650,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.